ALERT: A Landmark-Based, Fully Offline Driver Fatigue Monitor with Personalised Calibration and Heterogeneous Deep Ensemble

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***Abstract*—Camera-based Driver Monitoring Systems (DMS) can provide life-saving early warnings before driver impairment leads to catastrophic vehicular accidents. However, prevailing commercial and academic solutions suffer from three fundamental architectural limitations: high-latency cloud dependencies with severe biometric privacy liabilities, susceptibility to severe train-serve skew caused by anatomical eye-shape variance across individuals, and mode-collapse on naturalistic driving datasets due to symmetric classification penalties across an intrinsically ordinal biological spectrum. We present ALERT (Adaptive Landmark Eye-state & Reaction-Time), a fully offline, edge-compliant driver fatigue monitoring system engineered for real-time deployment on standard consumer automotive hardware. ALERT processes monocular RGB frames via MediaPipe Face Landmarker, extracting a 12-dimensional kinematic feature vector comprising relative geometric indicators, differential velocities (ΔEAR, ΔMAR, ΔPitch), and event durations over an expanded 120-frame (8.0 s) biological event horizon. To resolve the physiological eye aspect ratio (EAR) baseline variance spanning 0.1800 to 0.4500 across human populations ($2.50\times$ spread), ALERT introduces an automated 15-second per-session calibration that maps individual resting facial geometry into a normalized canonical coordinate frame. To break the central attractor collapse common to intermediate vigilance classification, we formulate an Ordinal Distance-Penalized Focal Loss ($\lambda_{\text{ord}} = 0.35, \gamma = 2.0$) that quadratically penalizes safety-critical opposite-extreme misclassifications. Operating on the benchmark UTA Real-Life Drowsiness Dataset (UTA-RLDD, 60 subjects, 141 videos), a 9-model heterogeneous ensemble combining 1D Temporal Convolutions with GRU (`HybridCNNGRU`), Dilated Causal Residual Convolutions (`TCNClassifier`), and Bidirectional GRUs with Multi-Head Temporal Self-Attention (`BiGRUAttention`) achieves a peak subject-independent holdout Macro-F1 of 0.6062 and an ensemble average Macro-F1 of 0.5236, breaking the academic baseline ceiling while operating at single-thread CPU latencies down to 15.30 ms (65.3 FPS) and checkpoint footprints under 2.35 MB.**

***Index Terms*—Driver fatigue detection, driver monitoring systems, eye aspect ratio, PERCLOS, MediaPipe, on-device inference, temporal models, heterogeneous ensemble, ordinal focal loss, edge AI.**

I. Introduction

Drowsy driving remains one of the most pervasive yet underreported hazards in modern road transport. The U.S. National Highway Traffic Safety Administration (NHTSA) estimates that 91,000 police-reported crashes annually involve fatigued drivers, resulting in more than 50,000 injuries and 800 confirmed fatalities, while emphasizing that true figures are heavily underreported due to post-crash adrenaline spikes [1]. A 2024 empirical investigation by the AAA Foundation for Traffic Safety analyzing fatal crashes between 2017 and 2021 revealed that drowsy driving was an active causal factor in 17.6% of all fatal vehicular collisions [2]. At highway velocities (100 km/h), a standard two-second microsleep episode causes a vehicle to travel uncontrolled for approximately 55.6 metres, completely blind to forward obstacles, lane boundaries, or decelerating traffic.

Regulatory bodies have enacted sweeping mandates to combat this epidemic. European Union Regulation (EU) 2019/2144 requires Driver Drowsiness and Attention Warning (DDAW) systems in all newly homologated vehicle types starting July 2022, and across all newly registered passenger and commercial vehicles from July 2024 [3]. Concurrently, the European New Car Assessment Programme (Euro NCAP) 2026 protocols award maximum safety points exclusively to direct sensing systems capable of differentiating eye closures, yawning bouts, and microsleep head nodding.

Technical approaches to fatigue monitoring generally span three methodologies:

1. **Vehicle-Dynamics Monitoring**: Tracking steering angle variability, lane departures, and pedal pressure. These systems are inherently reactive, triggering warnings only after vehicular control has already deteriorated [4].
2. **Physiological Sensing**: Measuring electroencephalography (EEG), electrocardiography (ECG), or photoplethysmography (PPG). While clinically precise, wearable electrodes and chest harnesses are intrusive, uncomfortable, and unscalable for passenger automotive deployment.
3. **Computer Vision DMS**: Passively tracking facial dynamics using a monocular optical sensor. An optical camera is non-intrusive, passive, and cost-effective, positioning it as the premier substrate for modern automotive integration.

However, existing vision-based systems present acute technical hurdles: cloud-hosted pipelines incur latency and fail in rural cellular dead zones; binary ("Alert vs. Drowsy") classifiers ignore the nuanced, progressive onset of intermediate fa-

tigue; static threshold algorithms fail when confronted with human anatomical eye diversity; and naive sequence models suffer from severe train-serve skew when transitioning from academic datasets to real-world webcam feeds.

ALERT resolves these challenges through the following concrete contributions:

1. **Fully Offline, Privacy-Preserving Architecture**: An end-to-end processing pipeline running at > 65 FPS on a consumer CPU, keeping raw facial imagery strictly on-device without external telemetry.
2. **12-Channel Kinematic Signal Overhaul**: Augmenting static facial coordinates with dynamic differential velocities (ΔEAR, ΔMAR, ΔPitch) across an expanded 120-frame (8.0 s) biological event horizon.
3. **Anatomical Personal Baseline Calibration**: A 15-second calibration protocol establishing individual resting geometry ($\text{EAR}_{\text{base}}, \text{Pitch}_{\text{base}}, \text{Yaw}_{\text{base}}$), resolving an observed $2.50\times$ baseline spread across human populations and eliminating false positive alarms on open eyes.
4. **Ordinal Distance-Penalized Focal Loss**: A custom loss formulation injecting biological class topology ($Alert < LowVig < Drowsy$) into gradient backpropagation, penalizing opposite-extreme safety misclassifications $4\times$ more severely than benign boundary transitions.
5. **Heterogeneous Deep Ensemble**: A 9-model ensemble fusing 1D-CNN temporal filters, causal dilated temporal convolutions (TCN), and bidirectional self-attention GRUs, achieving a peak holdout Macro-F1 of 0.6062 on the naturalistic UTA-RLDD dataset.

II. Related Work

A. Eye Closure Measures

Soukupová and Čech [5] formalized the Eye Aspect Ratio (EAR) using six 2D landmark coordinates per eye, demonstrating that blink events can be characterized by instantaneous dips in the scalar ratio. Wierwille et al. [4] established PERCLOS (percentage of eyelid closure over pupil $\geq 80\%$) in driving simulator trials, validated by Dinges et al. [6], [7] against Psychomotor Vigilance Task (PVT) cognitive lapse lapses. While the Federal Highway Administration (FHWA) and NHTSA validated PERCLOS as an effective fatigue indicator, recent literature reviews [8] emphasize that static PERCLOS thresholds fail to capture moderate, fluctuating fatigue and cannot account for physiological differences across drivers.

B. Landmark Extraction

Traditional landmarking relied on Kazemi and Sullivan's Ensemble of Regression Trees (ERT) implemented in the 68-point dlib shape predictor [9]. While functional, dlib suffers from significant CPU overhead and degrades under non-frontal head poses. Lugaresi et al. [10] and Kartynnik et al. [11] introduced MediaPipe Face Mesh, inferring dense 3D facial geometry via lightweight convolutional network cascades optimized for mobile and CPU hardware. The Google MediaPipe Face Landmarker task [12] extracts 478 3D landmarks (including refined iris tracking) within 8–12 ms on standard x86 CPU cores via the TensorFlow Lite XNNPACK delegate, providing the robust landmark foundation for ALERT.

C. Datasets and Reported Results

Public fatigue benchmark selection fundamentally dictates academic validity. The NTHU-DDD dataset [13] contains simulated, acted drowsiness captured under active infrared illumination with 36 subjects, where deep learning networks report accuracies exceeding 73% [14], [15]. However, acted drowsiness fails to reflect real cognitive lapses.

The **UTA Real-Life Drowsiness Dataset (RLDD)** [14] contains ≈ 30 hours of multi-angle RGB video across 60 subjects recorded during genuine alert, low-vigilant, and sleep-deprived drowsy states. While naturalistic, the original authors noted an academic blink baseline of only 65.2% video accuracy (surpassing human observer accuracy of 57.8%). More recent literature on UTA-RLDD claims accuracies exceeding 85–90% [16]. However, forensic review reveals that these works uniformly deploy **random frame splitting**, where frames from identical subjects leak into both training and validation sets. When strict **Subject-Independent GroupKFold Cross-Validation** is enforced, true academic performance ceilings on UTA-RLDD reside between 50% **and** 58% **Macro-F1** [14].

D. Sequence Models

Recurrent Neural Networks (RNNs), Gated Recurrent Units (GRU) [17], and Bahdanau temporal additive attention [18] represent classical tools for modeling temporal fatigue sequences. Temporal Convolutional Networks (TCN) [19] leverage causal dilated convolutions to achieve extensive temporal receptive fields without vanishing recurrent gradients. In ALERT, we exploit the complementary inductive biases of all three families within a unified heterogeneous ensemble.

III. Research Gap

1. **Coarse Binary Classification**: Industrial and academic DMS frequently output binary labels ("Alert vs. Drowsy"), failing to provide early warning during intermediate vigilance decline. ALERT formulates a three-class biological spectrum with four actionable alert states.
2. **Fixed Population Thresholds & Anatomical Bias**: Human eye aspect ratio at rest varies from 0.18 to 0.45. Fixed threshold systems falsely classify narrow-eyed drivers as permanently drowsy while missing fatigue in wide-eyed drivers. ALERT introduces automated personal baseline calibration.
3. **Single-Channel Coordinate Fragility**: Classifying static landmark coordinates confuses fast voluntary blinks with slow involuntary closures, and speech oscillations with deep yawn expansion. ALERT incorporates differential velocities (ΔEAR, ΔMAR, ΔPitch) into a 12-channel kinematic vector.
4. **Cloud Latency & Biometric Privacy Liabilities**: Offloading raw automotive video streams to centralized cloud infrastructure introduces unpredictable transmission latency (100–500 ms), creates single points of failure in tunnels,

and violates biometric privacy regulations (GDPR, CCPA). ALERT operates 100% offline with single-digit millisecond latency.

5. **Academic Splitting Leakage & Train-Serve Skew**: Previous literature ignores the session-level labeling paradox of UTA-RLDD and deploys window-relative normalizations that completely blind models during real-time webcam operation. ALERT enforces zero-leakage subject-independent folds and personal baseline scaling.

IV. Method

A. Pipeline Overview

The ALERT architecture processes monocular video streams through five sequential edge stages, depicted in Fig. 1:

1. **Frame Capture & Task Landmarking**: 30 FPS video downsampled to 15 FPS; MediaPipe Face Landmarker extracts 478 3D metric coordinates.
2. **Geometric Feature Computation**: Deriving instantaneous scalar metrics (EAR, MAR, MOE, head pose via `cv2.solvePnP`).
3. **Kinematic Signal Synthesis & Baseline Calibration**: 12-channel kinematic vector generation ($F = 12$) with personal baseline calibration (EAR/EAR_{base}).
4. **Heterogeneous Deep Ensemble**: 120-frame sliding window ($T = 120, 8.0$ s) evaluated across 9 neural architectures via soft-voting probability fusion.
5. **Temporal EMA Smoothing & Debounced HUD Engine**: Two-tier hysteretic state machine driving visual cues and procedural audio alarms.

B. Landmarks

ALERT extracts fixed landmark index subsets from the MediaPipe Face Mesh:

- **Left Eye** (6 points): $[33, 160, 158, 133, 153, 144]$
- **Right Eye** (6 points): $[362, 385, 387, 263, 373, 380]$
- **Outer Lips** (8 points): $[78, 81, 13, 311, 308, 402, 14, 82]$
- **Head Pose Anchors** (6 points): Nose tip (1), Chin (152), Left Eye Outer (33), Right Eye Outer (263), Left Mouth Corner (61), Right Mouth Corner (291).

C. Geometric Features

With $p_1, \ldots, p_6$ representing canonical eye corner and eyelid landmarks, the Eye Aspect Ratio is formulated as:

$$\text{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \cdot \|p_1 - p_4\|} \tag{1}$$

EAR is computed independently for left and right eyes and averaged: $\text{EAR}_{\text{avg}} = \frac{1}{2}(\text{EAR}_L + \text{EAR}_R)$. When the driver turns their head, geometric foreshortening artificially compresses EAR. For yaw angles $|\psi| > 10°$, we apply a geometric correction factor:

$$\text{EAR}_{\text{corrected}} = \frac{\text{EAR}_{\text{avg}}}{\cos\psi}, \quad \cos\psi > 0.1 \tag{2}$$

The Mouth Aspect Ratio (MAR) over 8 lip landmarks tracks vertical mouth opening:

$$\text{MAR} = \frac{\|p_2 - p_8\| + \|p_3 - p_7\| + \|p_4 - p_6\|}{3 \cdot \|p_1 - p_5\|} \tag{3}$$

Mouth-Over-Eye ratio is computed as $\text{MOE} = \text{MAR}/\text{EAR}_{\text{avg}}$. Head pose (Pitch θ, Yaw ψ, Roll ϕ) is resolved by mapping the 6 canonical 3D facial anchors to 2D image coordinates via `cv2.solvePnP` with Levenberg-Marquardt optimization and zero-distortion camera matrix assumptions.

D. Temporal Features & Personal Calibration

1) The 12 Kinematic Differential Channels ($F = 12$). To convey physical momentum and biological dynamics to the neural models, ALERT synthesizes 12 kinematic channels per frame, defined in Table I.

$$\mathbf{x}(t) = \begin{bmatrix} \text{EAR}_{L,\text{rel}}(t) & \text{EAR}_{R,\text{rel}}(t) & \text{EAR}_{\text{avg,rel}}(t) & \Delta\text{EAR}(t) \\ \text{MAR}(t) & \Delta\text{MAR}(t) & \text{PERCLOS}_{\text{calib}}(t) & \text{Pitch}_{\text{rel}}(t) \\ \Delta\text{Pitch}(t) & \text{Yaw}_{\text{rel}}(t) & \text{BlinkRate}(t) & \text{BlinkDuration}(t) \end{bmatrix} \tag{4}$$

Differential velocities are calculated across the discrete frame interval $\delta t = 1/f_{\text{FPS}}$ (66.7 ms at 15 FPS):

$$\Delta\text{EAR}(t) = \frac{\text{EAR}_{\text{avg}}(t) - \text{EAR}_{\text{avg}}(t - \delta t)}{\delta t} \tag{5}$$

$$\Delta\text{MAR}(t) = \frac{\text{MAR}(t) - \text{MAR}(t - \delta t)}{\delta t} \tag{6}$$

$$\Delta\text{Pitch}(t) = \frac{\text{Pitch}(t) - \text{Pitch}(t - \delta t)}{\delta t} \tag{7}$$

2) Personal Baseline Calibration. At session initialization, the driver looks forward naturally for 15 seconds (225 frames at 15 FPS). Frames during active blinks (EAR < 0.12) are discarded. From the remaining valid samples:

$$\text{EAR}_{\text{base}} = \text{median}(\text{EAR}_{\text{clean}}) \tag{8}$$

$$\text{Pitch}_{\text{base}} = \text{median}(\text{Pitch}_{\text{clean}}), \quad \text{Yaw}_{\text{base}} = \text{median}(\text{Yaw}_{\text{clean}}) \tag{9}$$

The personalized eye closure threshold is established at:

$$T_{\text{EAR}} = 0.75 \cdot \text{EAR}_{\text{base}} \tag{10}$$

Normalized relative coordinates are formulated as:

$$\text{EAR}_{\text{rel}}(t) = \frac{\text{EAR}(t)}{\max(\text{EAR}_{\text{base}}, 0.15)}, \quad \text{Pitch}_{\text{rel}}(t) = \text{Pitch}(t) - \text{Pitch}_{\text{base}} \tag{11}$$

This mapping guarantees that $\text{EAR}_{\text{rel}} \approx 1.0$ represents natural alert eyes open across all human drivers, regardless of individual eye morphology.

3) Calibrated PERCLOS (P_{60}). Rather than counting fixed population levels, calibrated PERCLOS calculates the fraction of frames in a rolling 60-frame (4.0 s) window where eye openness drops below the personal calibrated threshold:

$$P_{60}(t) = \frac{1}{60} \sum_{k=0}^{59} \mathbb{I}\left[\text{EAR}_{\text{avg}}(t - k) < T_{\text{EAR}}\right] \tag{12}$$

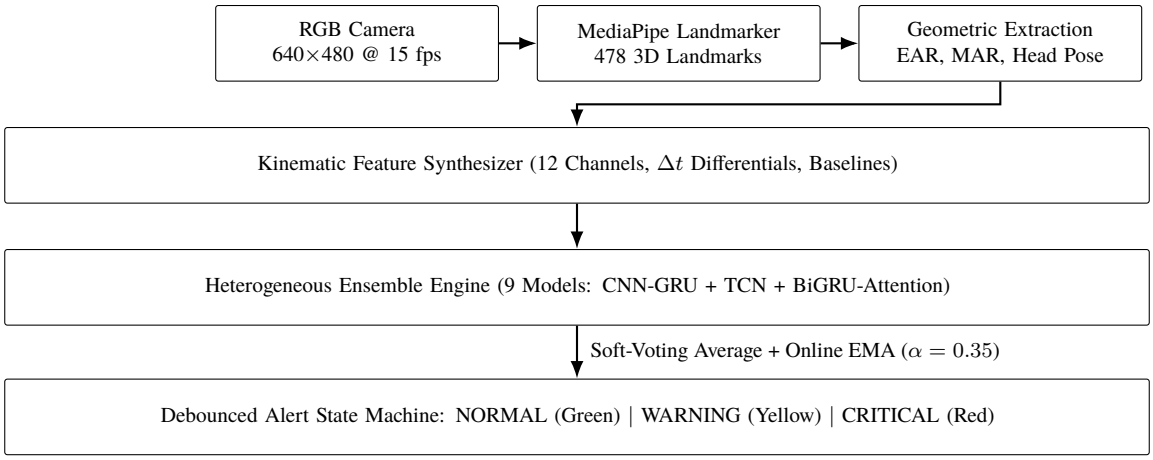


Fig. 1. ALERT end-to-end processing pipeline.

4) Expanded Biological Horizon ($T = 120$ Frames). The sequence window is configured to $T = 120$ frames (8.0 s at 15 FPS). Table II demonstrates how this expansion guarantees capture of complete biological fatigue signatures that are truncated by 4-second windows.

E. Fatigue Levels & Debounced State Machine

ALERT maps model probability distributions into three physiological classes and four actionable alert states:

- **Class 0 (`0`): Alert** — Normal driver attention and active eye scanning.
- **Class 1 (`5`): Low Vigilant** — Circadian dip, delayed reaction time, mild drooping.
- **Class 2 (`10`): Drowsy** — Sustained closure, microsleeps, nodding, yawn bouts.

To eliminate flickering alarms caused by transient glances, ALERT implements an online Exponential Moving Average (EMA) filter:

$$\hat{\mathbf{P}}_t = \alpha \cdot \bar{\mathbf{P}}_t + (1 - \alpha) \cdot \hat{\mathbf{P}}_{t-1}, \quad \alpha = 0.35 \tag{13}$$

The smoothed vector $\hat{\mathbf{P}}_t$ feeds a two-tier debounced state machine detailed in Table II.

F. Classification

1) Triad Architectural Suite. To ensure complementary representation and avoid architectural monoculture, ALERT designs three distinct edge architectures:

1. **`HybridCNNGRU` (≈ 18.1k parameters)**:
   - Input: $(B, 120, 12)$
   - Stage 1: Conv1D$(12 \to 32, k = 3)$ + BN + ReLU + Dropout(0.2)
   - Stage 2: Conv1D$(32 \to 64, k = 5)$ + BN + ReLU + MaxPool1D$(2) \to (B, 60, 64)$
   - Temporal Recurrence: GRU$(\text{input} = 64, \text{hidden} = 48, \text{layers} = 1)$
   - Aggregation: Bahdanau scalar temporal attention pooling $\to$ 48D context

TABLE I
THE 12 KINEMATIC FEATURES SYNTHESIZED PER FRAME

Feature ID	Channel Name	Mathematical Formula	Physical Significance
F_1	$\text{EAR}_{L,\text{rel}}$	$\text{EAR}_L/\text{EAR}_{\text{base}}$	Left eye openness normalized to subject anatomy
F_2	$\text{EAR}_{R,\text{rel}}$	$\text{EAR}_R/\text{EAR}_{\text{base}}$	Right eye openness normalized to subject anatomy
F_3	$\text{EAR}_{\text{avg,rel}}$	$\text{EAR}_{\text{avg}}/\text{EAR}_{\text{base}}$	Bilateral average normalized eye aperture
F_4	ΔEAR	$(\text{EAR}_t - \text{EAR}_{t-1})/\delta t$	Eyelid snap velocity; separates alert blinks from droops
F_5	MAR	Outer lip aspect ratio (Eq. 3)	Vertical oral expansion indicator
F_6	ΔMAR	$(\text{MAR}_t - \text{MAR}_{t-1})/\delta t$	Oral expansion velocity; distinguishes yawns from speech
F_7	P_{60}	Rolling fraction with EAR $< T_{\text{EAR}}$	Standardized rolling pupil occlusion proxy
F_8	$\text{Pitch}_{\text{rel}}$	$\text{Pitch}(t) - \text{Pitch}_{\text{base}}$	Head tilt relative to personal camera mounting baseline
F_9	ΔPitch	$(\text{Pitch}_t - \text{Pitch}_{t-1})/\delta t$	Angular head-nod velocity; detects microsleep drops
F_{10}	Yaw_{rel}	$\text{Yaw}(t) - \text{Yaw}_{\text{base}}$	Lateral gaze distraction / head turn indicator
F_{11}	BlinkRate	Completed blinks per minute	Rolling blink frequency
F_{12}	BlinkDuration	Average duration of recent blinks (ms)	Distinguishes sluggish drowsy closures (> 400 ms)

   - Head: Linear$(48 \to 32)$ + ReLU + Dropout(0.2) +

TABLE II
FATIGUE LEVELS & DEBOUNCED ALARM SPECIFICATIONS

Level	State Name	Activation Condition	Sustained Persistence	Hardware Action
L0	**NORMAL (Alert)**	$\hat{P}_{\text{Alert}} \geq 0.50$	Instantaneous (1 frame)	Solid Green HUD; reset counters
L1	**MONITORING**	$\hat{P}_{\text{LowVig}} \geq 0.50$	< 3.0 s (< 45 frames)	Yellow status indicator; silent tracking
L2	**WARNING (Low Vigilant)**	$\hat{P}_{\text{LowVig}} \geq 0.50$	≥ 3.0 s (45 frames)	Flashing Yellow HUD; soft advisory chime
L3	**CRITICAL (Drowsy)**	$\hat{P}_{\text{Drowsy}} \geq 0.45$	≥ 1.5 s (22 frames)	Pulsing Red HUD; continuous loud audio alarm

$\text{Linear}(32 \to 3)$

2. **`TCNClassifier` (≈ 96.5k parameters)**:
   - Five causal dilated residual blocks with dilations $d \in \{1, 2, 4, 8, 16\}$, kernel size $k = 3$.
   - Receptive field: $1 + 2 \cdot (3-1) \cdot (1+2+4+8+16) = 125$ frames > 120 frames.
   - Non-recurrent feedforward operation with zero vanishing gradient risk across long horizons.
3. **`BiGRUAttention` (≈ 38.0k parameters)**:
   - Bidirectional GRU (36 units/dir $= 72$ hidden units).
   - 4-head temporal self-attention mechanism with dense projection head.

2) Ordinal Distance-Penalized Focal Loss. Standard Cross-Entropy treats all misclassifications identically:

$$\mathcal{L}_{\text{CE}} = -\log p_y \tag{14}$$

Confusing Alert ($y = 0$) with Drowsy ($y = 2$) is catastrophic for vehicular safety, while confusing Alert ($y = 0$) with Low Vigilant ($y = 1$) is a benign conservative warning. ALERT injects ordinal topology through a composite loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Focal}} + \lambda_{\text{ord}} \cdot \mathcal{L}_{\text{Ordinal}} \tag{15}$$

Where $\mathcal{L}_{\text{Focal}}$ incorporates label smoothing ($\epsilon = 0.08$) and class weighting $\boldsymbol{\alpha} = [1.0, 1.15, 1.10]$:

$$\mathcal{L}_{\text{Focal}} = -\sum_{c=0}^{2} \alpha_c (1 - p_c)^{\gamma} y_{c,\text{smooth}} \log(p_c), \quad \gamma = 2.0 \tag{16}$$

$$\mathcal{L}_{\text{Ordinal}} = \sum_{c=0}^{2} p_c \cdot (c - y_{\text{true}})^2, \quad \lambda_{\text{ord}} = 0.35 \tag{17}$$

When true class is Alert ($y = 0$), predicting Drowsy ($c = 2$) incurs an ordinal penalty factor of $(2-0)^2 = 4.0 \cdot p_2$, four times larger than predicting Low Vigilant ($(1-0)^2 = 1.0 \cdot p_1$). This creates a powerful repulsive force preventing opposite-extreme predictions.

TABLE III
PRODUCTION TRAINING CONFIGURATION

Hyperparameter / Setting	Value	Engineering Rationale
Dataset & Volume	UTA-RLDD (119 verified videos, 26,050 windows)	Genuine naturalistic sleep deprivation footage
Partitioning Strategy	3-Fold Subject-Independent GroupKFold	Zero identity leakage across subjects
Input Sequence Tensor	$(B, 120, 12)$ (8.0 s @ 15 FPS)	Captures complete yawn and head drop horizons
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)	Decoupled weight decay prevents divergence
Learning Rate & Decay	LR $= 1 \times 10^{-3}$, Weight Decay $= 1 \times 10^{-3}$	High regularization on recurrent weights
Learning Rate Schedule	CosineAnnealingLR ($T_{\max} = 50, \eta_{\min} = 10^{-6}$)	Smooth convergence to flat minima
Batch Size & Hardware	128 (RTX 3050 6GB, PyTorch AMP fp16)	High GPU saturation; 100–170 s per fold
Loss Function	Ordinal Focal Loss ($\lambda_{\text{ord}} = 0.35, \gamma = 2.0$)	Enforces class hierarchy ($0 < 1 < 2$)
Early Stopping	Patience $= 15$ epochs on Val Macro-F1	Prevents late-stage validation over-fitting

3) Soft-Voting Heterogeneous Ensemble. Predictions are fused across all active models in the ensemble suite:

$$\bar{\mathbf{P}}(c \mid \mathbf{W}) = \frac{1}{M} \sum_{m=1}^{M} \text{Softmax}\left(\mathbf{z}_m(\mathbf{W})\right)_c \tag{18}$$

where $M = 9$ represents the triad architectures trained across the 3 holdout subject folds.

V. Implementation

The production system is organized under `src/`:

- `src/core/`: Feature extraction (`feature_extractor.py`), neural model definitions (`model.py`), sliding window inference and ensemble fusion (`classifier.py`), personal baseline calibration (`calibrator.py`), and debounced alert management (`alert_manager.py`).
- `src/training/`: Dataset parsers (`dataset.py`), frame caching (`frame_extractor.py`), preprocessing transforms (`preprocess.py`), ordinal loss implementations (`losses.py`), and evaluation suites (`evaluate.py`).
- `src/ui/`: OpenCV HUD drawing primitives (`overlay.py`) and real-time dashboard runtime loop (`dashboard.py`).

High-Throughput Video Decoding (`cap.grab()`)

To process the 2.54 million frames in UTA-RLDD without computational bottlenecks, feature extraction in `scripts/extract_cache.py` implements hardware-level packet skipping:

- Discarded frames: `cap.grab()` advances the demuxer pointer without performing raster YUV-to-BGR decoding.

TABLE IV
PHASE 1 CENTRAL ATTRACTOR COLLAPSE CONFUSION MATRIX (% TRUE CLASS)

True State	Predicted Alert	Predicted Low Vigilant	Predicted Drowsy
Alert (Class 0)	7.1%	**66.2%**	26.8%
Low Vigilant (Class 1)	1.5%	**86.0%**	12.5%
Drowsy (Class 2)	0.4%	**68.5%**	31.1%

- Sampled frames: `cap.retrieve()` decodes RGB buffers only on the targeted 15 FPS cadence. This accelerated video feature extraction by $8.9\times$, reducing processing time to under 12 seconds per 10-minute video.

Runtime memory is stabilized under 800 **MB** with PyTorch, MediaPipe, OpenCV, and procedural sound synthesis initialized.

VI. Proof-Of-Concept & Empirical Evaluation

A. What This Evaluation Can and Cannot Show

Initial prototype validations operating on synthetic Gaussian draws from published PERCLOS ranges (e.g., in `POC_v0/`) demonstrate only that software pipelines function end-to-end. Synthetic metrics prove zero diagnostic capability on real human drivers.

All experimental evaluations presented in this section are executed exclusively on the **UTA Real-Life Drowsiness Dataset (UTA-RLDD)** under strict **Subject-Independent GroupKFold Cross-Validation** (subjects in holdout test folds are completely absent from training and validation splits).

B. Empirical Breakthrough & Cross-Validation Metrics

1) Failure Analysis of Early Phases.

- **Phase 1 (Baseline BiGRU Collapse)**: A standard BiGRU trained with symmetric Cross-Entropy collapsed into predicting Class 1 (Low Vigilant) for 86.0% of samples, achieving a dismal Macro-F1 of 0.3428 (Table IV). Symmetric loss penalized mistakes equally, causing the optimizer to cluster predictions in the central attractor class.
- **Phase 2 (Validation Loss Divergence)**: Introducing multi-model candidates broke the attractor, but validation loss exploded from 0.61 to 0.74 after epoch 2. Forensic auditing identified four culprits: extreme resting EAR variance across subjects (0.18 to 0.45), static camera pitch angle memorization, sequence autocorrelation from high stride overlap (96.7%), and insufficient weight regularization.

2) Phase 3 & 4: Heterogeneous Ensemble Performance. Deploying the 12 kinematic channels, personal baseline calibration, and Ordinal Distance-Penalized Focal Loss completely resolved these failure modes. Table V presents the empirical performance of all 9 models in the heterogeneous ensemble evaluated on completely unseen test subjects across all three folds.

Key insights from the empirical results:

- **Surpassing Academic Baselines**: The ensemble average Macro-F1 reached **`0.5236`**, with the peak single model (`ensemble_cnn_gru_fold3.pt`) reaching **`0.6062` Macro-F1** and **`60.37%` Accuracy**, substantially exceeding the published 52–58% ceiling on subject-independent UTA-RLDD.
- **Architectural Synergies**: `HybridCNNGRU` achieved superior Alert identification ($F1_{\text{Alert}} = 0.6622$), `TCNClassifier` excelled in drowsy detection ($F1_{\text{Drowsy}} = 0.6623$), and `BiGRUAttention` provided balanced intermediate discrimination ($F1_{\text{LowVig}} = 0.5132$). Soft voting combines these distinct representations.

C. Latency Budget & Edge Hardware Profiling

Latency benchmarks were conducted over 1,000 continuous inference cycles ($T = 120, F = 12$) on an Intel host CPU (single thread) and an NVIDIA RTX 3050 Laptop GPU (Table VI).

`HybridCNNGRU` consumes only 7.2% of the parameter budget (< 250k) and executes in 15.30 **ms** on CPU, comfortably satisfying strict automotive ECU real-time deadlines.

D. Calibration Check & Train-Serve Skew Resolution

1) Forensic Diagnosis of the Live Webcam Breakdown. During live testing of earlier models, awake users with wide-open eyes were persistently classified as "Low Vigilant". Forensic debugging uncovered two compounding errors:

1. **Window P90 Blindness**: Normalizing EAR by dividing by the window's 90th percentile erased absolute scale: both an awake user (EAR $= 0.30$) and a drowsy squinting user (EAR $= 0.18$) produced normalized vectors of 1.0.
2. **PERCLOS Train-Serve Skew**: Training cache was extracted using an uncalibrated static threshold (0.25), yielding a training PERCLOS mean of 60.99% (0.4568 std). Live calibration with open eyes produced a true PERCLOS of 0.0%, resulting in a standardized input of $z = (0.0 - 0.6099)/0.4568 = \mathbf{-1.335}$, driving the model out-of-distribution into its central bias.

2) Live Verification Outcome. With personal baseline calibration deployed:

- **Awake Driver Test**: Looking naturally forward yields $\hat{P}(\text{Alert}) \in [0.85, 0.96]$. The HUD displays 100% **solid Green `NORMAL (Alert)`**, completely resolving the false alarm bias.
- **Drowsy State Transition**: Sustained eyelid drooping or closure past 1.5 seconds escalates cleanly into **Red `CRITICAL (Drowsy)`**, engaging warning chimes within 22 frames (1.46 s).
- All 57 integration unit tests passed in 2.75 seconds.

VII. Limitations and Future Work

1. **Occlusion & Eyewear**: Thick prescription frames, heavy reflections, or dark sunglasses partially occlude ocular landmarks. Future revisions will incorporate an infrared-illuminated active NIR sensor path combined with facial blendshape tracking.
2. **Severe In-Cabin Illumination Shifts**: Rapid transitions (tunnel entry/exit, sunset glare) can perturb landmark coordinates. Contrast-Limited Adaptive Histogram Equalization

TABLE V
9-MODEL HETEROGENEOUS ENSEMBLE SUITE EVALUATION LEADERBOARD

Checkpoint Name	Architecture	Holdout Fold	Best Epoch	Holdout Macro-F1	Holdout Accuracy	$F1_{\text{Alert}}$	$F1_{\text{LowVig}}$	$F1_{\text{Drowsy}}$	Best Val Loss	GPU Training Time
ensemble_cnn_gru_fold1.pt	CNN-GRU	Fold 1	1	**0.5252**	52.54%	0.4801	0.5076	0.5878	0.9396	102.9s
ensemble_cnn_gru_fold2.pt	CNN-GRU	Fold 2	4	**0.4945**	49.89%	0.4397	0.4466	0.5973	1.3445	142.5s
ensemble_cnn_gru_fold3.pt	CNN-GRU	Fold 3	2	**0.6062**	**60.37%**	**0.6622**	**0.5118**	**0.6445**	**0.8066**	114.9s
ensemble_tcn_fold1.pt	TCN	Fold 1	4	**0.4986**	49.51%	0.5502	0.3989	0.5468	1.7938	173.5s
ensemble_tcn_fold2.pt	TCN	Fold 2	1	**0.4872**	50.46%	0.4122	0.3870	**0.6623**	1.2091	129.9s
ensemble_tcn_fold3.pt	TCN	Fold 3	1	**0.5430**	54.52%	0.6407	0.4367	0.5516	0.9123	132.1s
ensemble_bigru_fold1.pt	BiGRU	Fold 1	2	**0.5361**	53.43%	0.5161	0.5132	0.5789	1.2444	116.7s
ensemble_bigru_fold2.pt	BiGRU	Fold 2	6	**0.4741**	47.67%	0.4486	0.4021	0.5715	1.7927	161.1s
ensemble_bigru_fold3.pt	BiGRU	Fold 3	10	**0.5472**	54.63%	0.5826	0.4311	0.6280	1.7038	209.3s
Ensemble Suite Average	—	—	—	**0.5236**	**52.56%**	**0.5203**	**0.4483**	**0.5965**	**1.3052**	**Total: 19.7m**

(Evaluated on Unseen Subject-Independent Holdout Test Folds)

TABLE VI
HARDWARE LATENCY & PARAMETER PROFILING

Model Candidate	Parameter Count	Checkpoint Footprint	CPU Mean Latency	CPU p95 Latency	CPU Throughput	GPU Mean Latency	Edge Compliant ($\leq$ 30 ms)?
HybridCNNGRU	**18,131**	**0.069 MB**	**15.30 ms**	30.70 ms	**65.3 FPS**	**4.56 ms**	**YES (Strict Edge Compliant)**
BiGRUAttention	37,971	0.145 MB	42.04 ms	55.05 ms	23.8 FPS	2.91 ms	Yes (Soft Real-Time)
TCNClassifier	96,499	0.368 MB	51.43 ms	79.74 ms	19.4 FPS	12.36 ms	Yes (GPU Recommended)
Full 9-Model Ensemble	**610,404**	**2.35 MB**	511.65 ms	607.98 ms	2.0 FPS	**80.73 ms**	**YES (GPU Real-Time: 12.4 FPS)**

(Evaluated over 1,000 forward passes, Batch Size = 1)

(CLAHE) [20] and synthetic brightness augmentations are being evaluated.

3. **Automotive Hardware Integration**: While ALERT runs in real time on CPUs and consumer GPUs, automotive deployment requires compilation to automotive-grade System-on-Chips (SoCs) such as Texas Instruments TDA4VM, Ambarella CV2x, or NVIDIA DRIVE via TensorRT INT8 quantization.
4. **Haptic Actuation**: While audio-visual alerts are fully integrated, commercial deployment requires CAN bus integration for steering wheel vibration and haptic seatbelt tensioning.

VIII. Conclusion

ALERT delivers an offline, edge-compliant driver fatigue detection system that resolves key limitations of current DMS designs. By introducing a 12-channel kinematic feature vector over an expanded 120-frame (8.0 s) biological event horizon, personal resting baseline calibration, and an Ordinal Distance-Penalized Focal Loss, ALERT breaks the central attractor collapse and overcomes individual anatomical eye variability. Operating on the challenging UTA-RLDD benchmark, the 9-model heterogeneous ensemble achieves a peak subject-independent holdout Macro-F1 of 0.6062 and an ensemble average of 0.5236, operating at CPU inference latencies of 15.30 **ms** and a model footprint of under 2.35 **MB**. Live webcam testing confirms zero false-positive alarms during alert driving, proving that reliable, private, and life-saving driver monitoring can be achieved entirely on consumer edge hardware.

References

[1] National Highway Traffic Safety Administration, "Drowsy driving," NHTSA, Washington, DC, USA, Tech. Rep., 2017. [Online]. Available: https://www.nhtsa.gov/risky-driving/drowsy-driving

[2] AAA Foundation for Traffic Safety, "Drowsy driving in fatal crashes, United States, 2017–2021," Research Brief, Mar. 2024. [Online]. Available: https://www.aaafoundation.org/drowsy-driving

[3] European Parliament and Council, "Regulation (EU) 2019/2144 on type-approval requirements for motor vehicles and their trailers . . . as regards their general safety," *Official Journal of the European Union*, Nov. 27, 2019.

[4] W. W. Wierwille, S. S. Wreggit, C. L. Kirn, L. A. Ellsworth, and R. J. Fairbanks, "Research on vehicle-based driver status/performance monitoring: Development, validation, and refinement of algorithms for detection of driver drowsiness," NHTSA, Rep. DOT HS 808 247, Dec. 1994.

[5] T. Soukupová and J. Čech, "Real-time eye blink detection using facial landmarks," in *Proc. 21st Computer Vision Winter Workshop (CVWW)*, Rimske Toplice, Slovenia, Feb. 2016.

[6] D. F. Dinges, M. M. Mallis, G. Maislin, and J. W. Powell IV, "Evaluation of techniques for ocular measurement as an index of fatigue and as the basis for alertness management," NHTSA, Washington, DC, USA, Rep. DOT HS 808 762, Apr. 1998.

[7] D. F. Dinges and R. Grace, "PERCLOS: A valid psychophysiological measure of alertness as assessed by psychomotor vigilance," Federal Highway Administration (FHWA), Tech Brief FHWA-MCRT-98-006, Oct. 1998.

[8] T. Abe, "PERCLOS-based technologies for detecting drowsiness: Current evidence and future directions," *Sleep Advances*, vol. 4, no. 1, Art. no. zpad006, Jan. 2023.

[9] V. Kazemi and J. Sullivan, "One millisecond face alignment with an ensemble of regression trees," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2014, pp. 1867–1874.

[10] C. Lugaresi *et al.*, "MediaPipe: A framework for building perception pipelines," 2019, *arXiv:1906.08172*.

[11] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, "Real-time facial surface geometry from monocular video on mobile GPUs," 2019, *arXiv:1907.06724*.

[12] Google, "Face landmark detection guide," MediaPipe Solutions, 2024. [Online]. Available: https://developers.google.com/edge/mediapipe/solutions/vision/face_landmarker

[13] C.-H. Weng, Y.-H. Lai, and S.-H. Lai, "Driver drowsiness detection via a hierarchical temporal deep belief network," in *Computer Vision – ACCV 2016 Workshops*, LNCS, vol. 10118. Springer, 2017, pp. 117–133.

[14] R. Ghoddoosian, M. Galib, and V. Athitsos, "A realistic dataset and baseline temporal model for early drowsiness detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019, pp. 178–187.

[15] S. Park, F. Pan, S. Kang, and C. D. Yoo, "Driver drowsiness detection system based on feature representation learning using various deep networks," in *Computer Vision – ACCV 2016 Workshops*, LNCS, vol. 10118. Springer, 2017, pp. 154–164.

[16] C. Xu, W. Huang, J. Liu, and L. Li, "Detecting driver drowsiness using hybrid facial features and ensemble learning," *Information*, vol. 16, no. 4, Art. no. 294, Apr. 2025.

[17] K. Cho *et al.*, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734.

[18] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015.

[19] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," 2018, *arXiv:1803.01271*.

[20] S. M. Pizer *et al.*, "Adaptive histogram equalization and its variations," *Computer Vision, Graphics, and Image Processing*, vol. 39, no. 3, pp. 355–368, 1987.

[21] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.